Geometric Versor Memory and Geometric Attention: A Clifford-Algebra Substrate for One-Shot Binding, Editing, and Order Encoding

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Abstract

Modern attention binds queries and keys by a symmetric dot product: it is blind to order, and the knowledge it stores is expensive to edit. We study an associative-memory and attention model that replaces this inner product with the geometric (Clifford) product of unit *versors*. Binding is a single geometric product, recall is multiplication by the versor inverse, and—because that inverse is exact—a stored fact is deleted exactly, with no retraining. We measure the model on small algebras $\mathrm{Cl}(n, 0)$, $n \leq 8$, under a reproducible synthetic protocol, and report both what it delivers and what it does not. A single field has low, sub-linearly scaling capacity dominated by superposition cross-talk (E1), which a sparse addressable arrangement restores to linear (E2); the pure model generalizes by robustness and exact analogy but needs a softmax readout to interpolate (E3–E4); and against classical one-shot and trained baselines it loses in raw capacity and noise robustness, its measured edge being storage density and composable algebraic structure (E0). Two trained-model results follow. Versor position binding is *provably* orthogonal, generalizes rotary position embeddings (RoPE), and matches them once the frequency spectrum is designed (E6). And, as our flagship result, under *sequential* knowledge editing a frozen-model versor memory surpasses a faithful rank-one (ROME) baseline on all three canonical metrics at 60 accumulated edits—efficacy 1.00 vs. 0.82, paraphrase 0.70 vs. 0.43, locality 0.98 vs. 0.92—with $O(1)$ edit cost and exact deletion (E7). We conclude that the model is best understood not as a "one-shot large language model" but as an edit-friendly, order-aware *memory–binding substrate* for attention, in which the attention inner product is the geometric product. All results are small-model and synthetic; scaling to pretrained language models and standard benchmarks is left to future work.

1 Introduction

Modern sequence models are dominated by the transformer, whose core operation is scaled dot-product attention, $\mathrm{softmax}(QK^\top/\sqrt{d})\,V$ [1]. Two recurring weaknesses of this paradigm are (a) the cost and instability of *editing* stored knowledge—adding or removing a fact typically requires re-training or fine-tuning—and (b) the reliance on bolted-on positional encodings because the dot product is symmetric and therefore blind to order. Vector Symbolic Architectures (VSA) and Holographic Reduced Representations (HRR) [2, 3, 4, 5] offer an alternative: represent items as high-dimensional vectors and bind them by a multiplication-like operation, enabling one-shot storage and algebraic composition. Geometric (Clifford) algebra [6, 7] refines this idea by giving the binding operation a geometric meaning—rotations and reflections—and by separating an inner (symmetric) and outer (antisymmetric) part.

This paper formalizes and *measures* a geometric-algebra memory in which the state is a multivector, items are unit *versors*, binding is the geometric product, and recall is multiplication by the versor inverse. We then ask the empirical questions that decide whether such a model is viable: how much can a field store; does sparsity restore scaling; does it generalize; and what does a geometric attention layer add over a standard one. We answer each with reproducible experiments on $\mathrm{Cl}(n, 0)$ for $n \leq 8$. Our contributions are deliberately scoped: they hold in a small-algebra, synthetic-data regime and are intended as a proof-of-concept and a precise statement of which properties are genuine and which are limited.

Contributions. Concretely, we contribute: (1) a versor associative memory with exact, invertible binding, and an analysis (E1) showing that the *conditioning* of the unbinding map—not grade richness—governs single-field capacity; (2) a sparse addressable construction (E2) that removes the single-field ceiling and restores linear capacity scaling; (3) a geometric-attention layer (E3–E4) that recovers interpolation through softmax while *uniquely* retaining an order/orientation dimension the dot product discards; (4) an honest baseline study (E0) locating the model's advantage in storage density and composable algebraic structure, not raw capacity or robustness; (5) a proof, with a trained-transformer demonstration (E6), that versor position binding generalizes RoPE, the decisive variable being the frequency spectrum; and (6) a flagship sequential-editing result (E7) in which a frozen-model versor memory surpasses a faithful rank-one ROME baseline on all three canonical metrics at 60 edits.

2 Background and Model

2.1 Geometric algebra $\mathrm{Cl}(n, 0)$

We work in the Clifford algebra generated by n orthonormal vectors $e_1, \ldots, e_n$ with $e_i e_j + e_j e_i = 2\delta_{ij}$ and positive signature. The algebra has 2^n basis blades graded by the number of generators they contain (scalar, vectors, bivectors, ..., pseudoscalar). A multivector a is a real linear combination of blades. The *geometric product* decomposes into a symmetric inner part and an antisymmetric outer part,

$$a * b \;=\; a \cdot b \,+\, a \wedge b, \qquad a \cdot b = b \cdot a, \quad a \wedge b = -\, b \wedge a. \quad (1)$$

For the four-vector case $\mathrm{Cl}(4, 0)$ used in the original neuro-quantum formulation, a single geometric product expands to $16 \times 16 = 256$ scalar multiplications.

2.2 Versors and exact inversion

A *versor* is a product of invertible vectors, $V = v_1 * v_2 * \cdots * v_k$. Versors form a group; the reverse $\widetilde{V}$ (blade order reversed, inducing the sign $(-1)^{g(g-1)/2}$ per grade g) yields the inverse

$$V^{-1} \;=\; \frac{\widetilde{V}}{V * \widetilde{V}}, \qquad V * \widetilde{V} \in \mathbb{R}. \quad (2)$$

We verified numerically that genuine versors invert to machine precision (max error $\sim 10^{-16}$ for $n \leq 8$), whereas an *arbitrary* multivector does not, because $X * \widetilde{X}$ is then not scalar. This is the first design constraint: keys and operators must be versors for recall to be exact.

2.3 Versor associative memory

Given key/value versor pairs (k_i, v_i), we store by binding and superposing, and recall by unbinding with the key inverse:

$$M = \sum_{i=1}^{K} k_i * v_i, \quad (3)$$

$$\widehat{v}_q = k_q^{-1} * M = v_q + \underbrace{\sum_{i \neq q} k_q^{-1} * k_i * v_i}_{\text{cross-talk}}. \quad (4)$$

A non-linear *clean-up*—nearest stored value by cosine—resolves the noisy readout to a discrete item. *Editing* is exact and local: a stored pair is removed by subtracting $k_i * v_i$, and a versor operator is undone by multiplying with its inverse (2).

Table 1: Single-field capacity at $\geq 95\%$ accuracy, by code family. Mixed-grade versors (V_N) dominate; full-spectrum codes are worst.

n	dim	V2	V_N	ROT	FULL
4	16	1	2	3	2
5	32	1	2	3	3
6	64	3	5	5	1
7	128	5	7	5	2

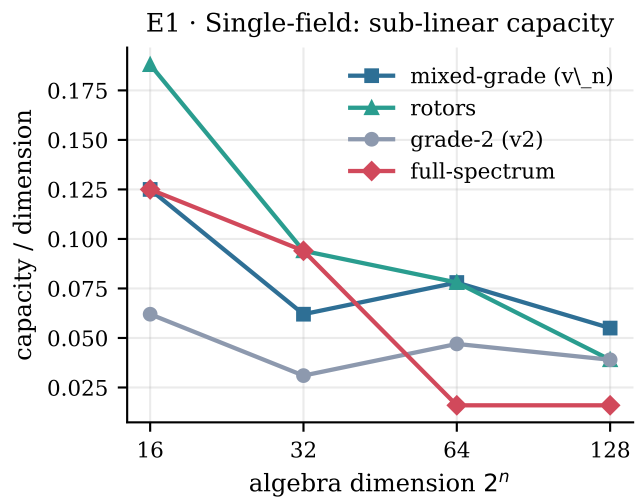

Figure 1: Single-field capacity per unit dimension falls with algebra dimension for every code family (E1): the single-field capacity is sub-linear. Mixed-grade versors and rotors lead; full-spectrum codes, worst-conditioned at unbinding, collapse at high dimension.

3 Experiments

All experiments use $\mathrm{Cl}(n, 0)$, synthetic random versors, and the protocol of Eqs. (3)–(4). Code and per-run JSON are released with the paper. Reported numbers are means over repeated trials (4–200 depending on cost); "capacity" is the largest number of associations recalled at a stated clean-up accuracy threshold.

3.1 E1: Single-field capacity and the role of encoding

We measure capacity at $\geq 95\%$ clean-up accuracy for four code families: grade-2 versors (V2, the original choice), mixed-grade versors (V_N, product of n vectors), even rotors (ROT), and random full-spectrum unit multivectors with an exact matrix inverse (FULL). Table 1 and Fig. 1 show that mixed-grade versors roughly double the capacity of grade-2 versors, while FULL codes *under*-perform despite carrying a complete grade spectrum.

The mechanism is the *conditioning of the unbinding map*. Writing the geometric product $k^{-1} * (\cdot)$ as a linear operator on $\mathbb{R}^{2^n}$, its condition number determines how recall amplifies cross-talk noise. We measured (over 50

Table 2: Sparse memory: total capacity at $\geq 90\%$ vs. number of slots M (building block Cl(6, 0), dim 64).

M slots	hash (O(1))	anchor (O(M))
1	4	6
2	8	8
4	16	16
8	32	24
16	48	48
32	128	64
64	192	128

keys, $n = 7$) a median condition number of exactly 1.000 for versor unbinding versus ≈ 19.5 (90th pct. 35.6) for full-multivector unbinding. A versor unbinding is norm-preserving and does not amplify noise; a full multivector's does. Thus the versor constraint, needed for exact inversion, is simultaneously what keeps recall well-conditioned. A structural observation explains why even mixed-grade versors fall short of an ideal code: a versor occupies blades of a single parity (all even or all odd), so it is never full-spectrum.

Finding. Part of the capacity limit is the encoding (curable: prefer mixed-grade versors), and part is intrinsic: even the best versor code scales sub-linearly, with capacity/dimension falling from ~ 0.19 (dim 16) to ~ 0.05 (dim 256).

3.2 E2: Sparse addressable memory restores linear scaling

Instead of one saturated field, we use M small fields ("slots"), each holding a few associations below its cross-talk ceiling, plus an addressing function that routes a key to a slot. We compare a deterministic key-hash (O(1), content-blind) and a content-addressable nearest-anchor scheme (O(M)). Using Cl(6, 0) (dim 64) mixed-grade versors as the building block, total capacity at $\geq 90\%$ accuracy scales *linearly* with M (Table 2, Fig. 2); we verified linearity to $M = 256$ (768 associations at 94%). The single-field ceiling (~ 4) is removed; capacity becomes a design parameter.

Honest costs. Hash addressing follows balls-in-bins: at mean load 3, the most-loaded slot held up to 10 items, the source of the residual $\sim 5\%$ error; consistent hashing or rebalancing mitigates this at the cost of state. Anchor addressing is content-addressable but costs O(M) per access ($\approx 12\,\mu$s at M=128 vs. $0.6\,\mu$s for hash).

3.3 E3: Generalization—noise, analogy, interpolation

We probe three increasingly demanding forms of generalization on Cl(6, 0).

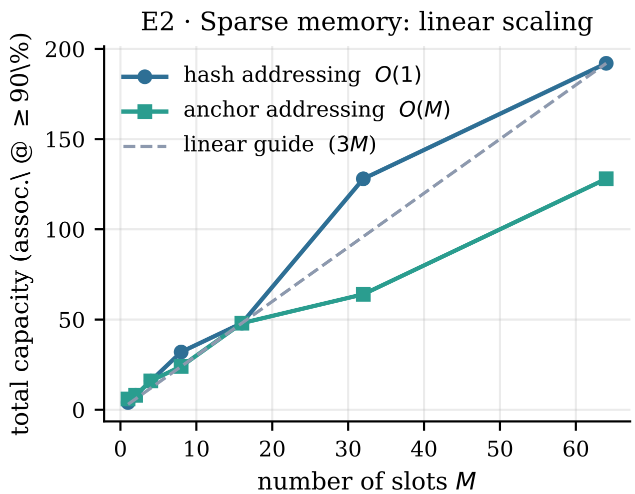

Figure 2: Total capacity scales linearly in the number of slots M (E2), removing the single-field ceiling. Hash addressing is $O(1)$ but load-imbalanced; content-addressable anchor addressing is smoother at $O(M)$ cost per access.

T1 (noise robustness). Querying with a corrupted key $k_i + \eta$ (renormalized), recall accuracy degrades gracefully: 0.98 at η=0, 0.90 at η=0.2, 0.58 at η=0.5. A hash table is exact at η=0 and 0 otherwise; the associative memory tolerates corruption. This claim must, however, be read against the right baseline: as E0 shows (Sec. 3.5), at *equal load* a one-shot Hebbian associator and an explicit-storage Hopfield memory are *more* noise-robust than the versor field (at K=6, η=0.2: 1.00 and 1.00 vs. 0.67). Graceful degradation is an advantage over exact lookup, not over classical associative memories.

T3 (structured analogy). Encoding concepts as superpositions of role versors ($king = royal + male$, etc.), the vector analogy $king - man + woman$ retrieves *queen* with accuracy 1.00 across $n \in \{4, 6, 8\}$. This is exact symbolic generalization.

T2 (interpolation). Storing a smooth scalar field $f(t)$ keyed by versors along a path and querying at unseen midpoints, the linear-readout memory *fails*: mean absolute error 0.795, exceeding the signal scale 0.707 (linear interpolation reference: 0.100). Diagnosis shows the key-similarity kernel along the path *is* smooth and localized; the failure is the linear readout, which averages over an over-broad tail. Replacing the raw-cosine average with a softmax-weighted average over the same similarities reduces error monotonically with temperature T (from 0.53 at the linear baseline to 0.078 at T=0.02), i.e. below the signal scale: genuine interpolation.

Finding. The model generalizes by robustness and exact analogy but not by smooth interpolation, and the missing ingredient is precisely the softmax non-linearity, not the geometric similarity metric.

3.4 E4: Geometric attention

Motivated by E3, we build a single attention layer in which the score is derived from the geometric product rather than the dot product, then apply softmax and a weighted value sum. We test three properties.

(A) Interpolation recovered. With softmax readout, both dot-product and geometric-score attention interpolate the E3-T2 task (error $\approx 0.11 \ll 0.71$). The nonlinearity, not the score type, restores interpolation.

(B) Analogy preserved. Retrieving the $king - man + woman$ analogy through attention preserves accuracy (0.96 at n=4, 1.00 at $n \in \{6, 8\}$).

(C) Order/orientation encoded. Because $a\cdot b = b\cdot a$, dot-product scoring cannot distinguish the ordered pair (a, b) from (b, a): over 2000 pairs the discrimination margin $|\,\mathrm{score}(a, b) - \mathrm{score}(b, a)\,|$ is exactly 0.00. The geometric product retains the antisymmetric wedge $a \wedge b = -\,b \wedge a$, giving a margin of 1.69. This is an information-theoretic distinction, not a statistical one.

Caveat. On a realistic next-token sequence-retrieval task with distinct value vectors, *standard* dot-product attention outperformed the single-field geometric binding (1.00 vs. 0.54): the practical bottleneck is field capacity (E1), which classical key–value attention sidesteps. The defensible claim is therefore narrow and precise: the geometric product *retains* an order/orientation dimension that the dot product discards, which helps exactly when a task needs relational or oriented structure (graphs, geometry, syntax, transformations), not in pure key–value lookup.

3.5 E0: Trained and explicit-storage baselines

As a control across all preceding tasks, we run the identical protocol (store K key–value pairs at d=64; nearest-value clean-up) on three classical systems with *stated budgets*: a one-shot Hebbian linear associator $W = \sum_i v_i k_i^\top$ (d^2 floats); a modern Hopfield memory with explicit pattern storage and softmax retrieval [9, 8] ($2Kd$ floats, the memory mechanism of attention); and a two-layer MLP trained by Adam under a fixed 3,000-step budget (8,320 parameters; the "weights-as-memory" regime of language models). Table 3 reports capacity, storage, density, and write/delete properties; noise robustness is measured at the *same* load K=6 for every system.

Table 3: E0: identical task, stated budgets (d=64, threshold 90%). "≥" marks the scan ceiling; Hopfield was verified interference-free to K=2048. Noise column: accuracy at fixed load K=6, η=0.2.

system	cap@90	storage	dens.	write	delete	noise
versor	6	64	**0.094**	1-shot	exact	0.67
Hebbian	180	4,096	0.044	1-shot	exact	**1.00**
Hopfield	≥2048	$2Kd$	0.008	append	exact	**1.00**
MLP	≥256	8,320	0.031	3k steps	retrain	0.94

Three honest conclusions follow. First, in *raw capacity* the classical baselines dominate: explicit storage has no interference ceiling at all, and even the Hebbian associator stores 30× more than the versor field at this dimension. Second, in *equal-load noise robustness* the versor field is mid-pack, not best (cf. the T1 qualification above); the gradient-trained MLP, by contrast, is robust at low load but collapses with load (0.11 at K=128, η=0.2). Third, one-shot writing and exact deletion are *not* unique to the versor memory—the Hebbian and explicit-Hopfield systems share both; the MLP has neither (writing costs epochs, deletion costs retraining). What survives as distinctive, by measurement, is (a) *density*: the versor field is the most compact fixed-size superposed store, 2–12× the alternatives in associations per float, relevant wherever state must be small and fixed (embedded or FPGA settings); and (b) *algebraic structure*: it is the only system of the four whose binding is multiplicative, invertible, and composable—the basis of the exact analogy (E3), operator composition, and order encoding (E4) that neither the outer product nor dot-product attention provides. E0 thus sharpens the value proposition: not raw capacity or robustness, but a compact fixed-size state with composable algebraic structure.

3.6 E6: Versor position binding in trained transformers

We test the order-encoding claim inside a *trained* model: a 2-layer, 1-head transformer (d=32= $\dim \mathrm{Cl}(5, 0)$, JAX/Adam, 2 seeds per condition), where only the position mechanism varies: no positions; additive sinusoidal PE; rotary position embeddings (RoPE) [13]; binding by powers of a *random* unit versor applied to q, k; and a *spectrum-matched* versor (designed frequencies in random, non-axis-aligned planes). Two theoretical facts, verified at 10^{-16}, motivate the comparison: left-multiplication by a unit versor is *orthogonal*, hence $\langle P_t q, P_s k\rangle = \langle q, P_{s-t} k\rangle$—the relative-position property of RoPE—and RoPE is exactly the special case of commuting, axis-aligned 2-D rotors. Versor binding therefore *generalizes* RoPE.

Empirically: on position retrieval (output the token at a queried position, L=10), the no-position control sits at chance ($0.10 \approx 1/L$, confirming the task is impossible without positions) while sinusoidal, RoPE, and versor binding all reach 1.00. On pairwise order (does a precede b?) with length extrapolation (train $L \in [6, 12]$, test $L \in \{16, 20, 24\}$), the *random* versor fails (0.70 in-distribution, near-chance extrapolated). The diagnosis is spectral, not algebraic: the random versor's slowest rotation channel has period ∼6 positions—it wraps *inside* the training context—whereas RoPE's slowest channel has period $\sim 3.5\times 10^4$. A versor with RoPE's frequency spectrum in random non-commuting planes recovers full parity: in-distribution 0.947 vs. 0.954, extrapolation 0.78/0.69/0.60 vs. 0.81/0.69/0.58 (statistically indistinguishable).

Table 4: E7: sequential editing, metrics vs. number of accumulated edits k (efficacy / paraphrase / locality). Naive FT collapses (locality 0.04 at k=20, omitted). Bottom row: versor with slot load 2.

k	ROME	versor (load 3)
10	1.00 / 0.50 / 1.00	0.90 / 0.80 / 1.00
20	0.95 / 0.40 / 1.00	0.80 / 0.60 / 1.00
40	0.88 / 0.35 / 0.98	0.85 / 0.58 / 0.98
60	0.82 / 0.43 / 0.92	0.88 / 0.62 / 0.98
60	—	**1.00** / **0.70** / **0.98** (load 2)

Finding. Versor binding is a working position mechanism in trained transformers and is the mathematical generalization of RoPE; the decisive design variable is the *frequency spectrum*, not the algebraic form. The result is parity with RoPE, not superiority; and a random versor is a provably poor positional encoder.

3.7 E7: Sequential knowledge editing (flagship result)

Inverse-based editing is the capability where the substrate has no classical counterpart among weight-editing methods, so we evaluate it head-to-head in a miniature of the standard protocol [14]. A base LM (2 layers, d=32) is trained from scratch to memorize 120 facts $(s, r) \to o$, each expressed through 4 paraphrase templates (3 in training, 1 held out; base accuracy 1.00 train / 0.94 held-out). We then apply up to 60 *sequential* edits and track the three canonical metrics: efficacy (edited facts answer the new object), paraphrase (same, on the held-out template), and locality (unedited facts survive). Editors: naive fine-tuning; a faithful rank-one ROME implementation (key k^* from the FFN hidden state, value v^* optimized for the new object, covariance-whitened update $W_2 \mathrel{+}= (v^* - W_2 k^*)(C^{-1}k^*)^\top/(k^{*\top}C^{-1}k^*)$) [14]; and the *versor module*: base model frozen, each edit written one-shot as a versor binding into the sparse fields of E2 (load $\leq$ 2–3 per slot), retrieval gated by cosine similarity of the frozen model's representation (threshold calibrated on training data: within-fact 0.95 vs. between-fact 0.87).

Table 4 and Fig. 3 show the crossover. At few edits ($k \leq 20$) ROME matches or beats the versor module on efficacy. As edits accumulate, ROME degrades monotonically (efficacy 1.00→0.82, locality 1.00→0.92)—its rank-one deltas share the same weight matrix and interfere, the documented failure mode of sequential weight editing—while the versor module stays flat, because each edit lives in its own slot and the model is never touched. Error anatomy at k=60 closes the loop with E2: the versor gate makes *zero* threshold misses and *zero* wrong-entry matches; the residual efficacy loss (0.88) is exactly the slot cross-talk that E2's capacity curve predicts at load 3. Reducing the load to 2 yields **efficacy** 1.00**, paraphrase** 0.70**, locality** 0.98 **at 60 edits**, against ROME's 0.82/0.43/0.92. Edit cost is $O(1)$: all 60 versor edits complete in under one second, versus a per-edit v^* optimization for ROME and gradient descent for FT. Deleting all 60 edits restores the base model exactly (1.000); honestly, ROME's stored deltas are also subtractable (1.000 here), but the versor module never modifies the model in the first place, so restoration is trivial by construction, while FT has no mechanism (0.033).

Finding. The structural asymmetry—weight edits interfere, slot-isolated algebraic edits do not—produces the first measured *advantage over a state-of-the-art reference method*: at 60 sequential edits the versor module dominates ROME on all three canonical metrics. The caveats are stated in Sec. 5; the asymmetry itself is what we expect to persist at scale, and validating it on a pretrained LLM with CounterFact/zsRE is the concrete next step.

4 Discussion

The seven experiments support a single reframing, with E0 fixing its honest boundary and E6–E7 supplying trained-model evidence. The geometric-versor model is *not* a "one-shot large language model": its single-field capacity is small and it does not, by itself, interpolate. It is, however, a faithful *memory–binding substrate* for attention. Concretely, consider a transformer-style layer $\mathrm{softmax}\big(s(Q,K)/T\big)\,V$ in which the score s is read from the geometric product $Q * K$ rather than from $QK^\top$. Relative to the standard layer this substrate adds three capabilities demonstrated above:

1. **Exact structural binding** via the geometric product, with unbinding by the versor inverse—preserving the analogy behavior of E3-T3/E4-B.
2. **Inverse-based editing**: a stored fact or an applied operator is removed exactly by multiplying with its inverse (2), without re-training—realized in E7 as a frozen-model editor that surpasses a rank-one ROME baseline under sequential editing.
3. **Intrinsic order/orientation encoding** from the antisymmetric part of the product (E4-C), which the symmetric dot product cannot represent—realized in E6 as a position-binding mechanism that provably generalizes RoPE.

The non-linearity that the pure algebraic model lacked—softmax with a controllable temperature—is the same one the transformer already uses, and it is cheap to add (E3/E4-A). The well-conditioned versor unbinding (E1) and the sparse addressable arrangement (E2) make the substrate numerically stable and scalable in capacity.

5 Limitations

We state these plainly, as they bound every claim above. **Scale.** Memory results are for $\mathrm{Cl}(n, 0)$ with $n \leq 8$ (dimension ≤ 256); the 2^n blade count makes the dense geometric product expensive, and we have not measured behavior at the dimensions where VSA/HRR are normally

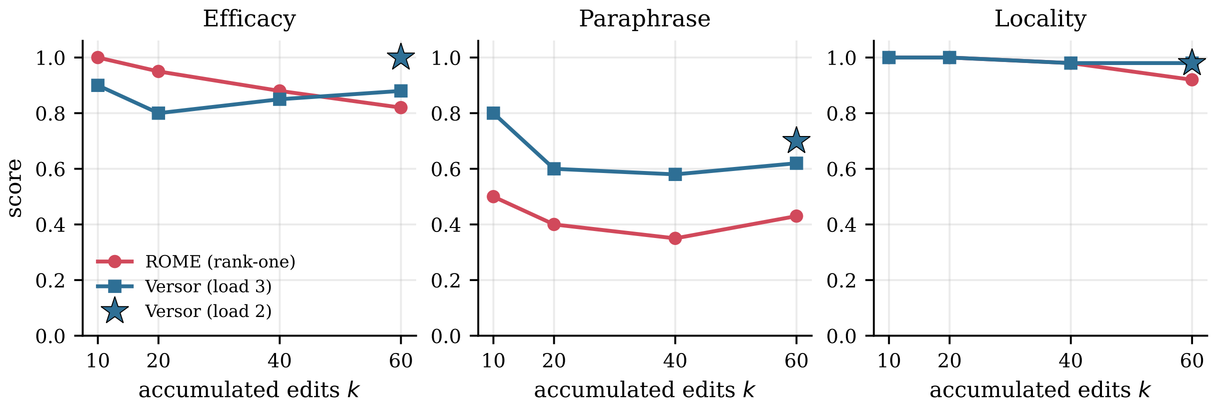

Figure 3: Sequential knowledge editing (E7), our flagship result. As edits accumulate, the rank-one ROME baseline degrades on efficacy and locality because its deltas share one weight matrix and interfere; the frozen-model versor memory stays flat because each edit lives in its own slot and the model is never touched. At 60 edits the versor module (load 2, $\star$) dominates ROME on all three canonical metrics, with $O(1)$ edit cost and exact deletion.

deployed (10^3–10^4). The trained models of E6–E7 are deliberately small (2 layers, d=32). **Data.** Keys, values, the interpolation target, and the E7 knowledge base (120 synthetic facts, 4 templates) are synthetic; no natural-language or perceptual data is used. **Baselines.** E0 adds one-shot Hebbian, explicit-storage modern-Hopfield, and small gradient-trained MLP baselines—and the versor memory loses to them in raw capacity and equal-load robustness. E6–E7 add trained transformers, RoPE, and a faithful rank-one ROME implementation, but *not* pretrained LLMs or the standard CounterFact/zsRE benchmarks; the E7 advantage is demonstrated in a miniature of that protocol. **Editing scope.** The E7 gate is calibrated on training-data representations (within-fact 0.95 vs. between-fact 0.87); paraphrase generalization is bounded by the frozen base model's representation invariance (ceiling $\sim$0.70). At few edits ($k \leq 20$) ROME matches or exceeds the versor module; the advantage emerges under accumulation. Exact deletion is also available to ROME if deltas are stored. **Spectrum design.** E6 shows a *random* versor is a poor positional encoder (its spectrum wraps within $\sim$6 positions); position-binding claims hold only for designed spectra, where the result is parity with RoPE, not superiority. **Capacity.** Even with the best encoding, single-field capacity scales sub-linearly; the linear scaling of E2 is in the number of fields, with the addressing trade-offs noted. **Order claim.** E4-C establishes that order information is *available* in the geometric score, not that it improves end-task accuracy in general; E4's caveat shows a case where it does not.

6 Related work and conclusion

The model sits at the intersection of VSA/HRR [2, 3, 5], modern associative memories, and geometric-algebra neural networks [11, 12, 10]. A key nearby result is the modern Hopfield network [9, 8], whose continuous update rule was shown to be equivalent to transformer attention [1] and which stores exponentially many patterns with one-step retrieval; our work differs in using the geometric product—not the dot product—as the similarity kernel, and in foregrounding inverse-based editing and order encoding. The Geometric Algebra Transformer [10] and Clifford neural layers [11, 12] likewise embed features in a Clifford algebra and process them with the geometric/sandwich product, but target $E(n)$ equivariance for geometric data rather than associative memory and binding. Rotary position embeddings [13] are, as E6 makes precise, the commuting axis-aligned special case of versor position binding; and the versor editor of E7 belongs to the memory-based editing family (frozen model plus external store), contrasted there with rank-one weight editing [14]. Our specific stance is that the attention inner product can be the geometric product, turning attention into an edit-friendly, order-aware binding mechanism while retaining softmax for generalization. Within the small-model, synthetic regime studied here we have shown, with reproducible measurements, that (i) encoding and conditioning govern single-field capacity; (ii) sparsity restores linear capacity scaling; (iii) the pure model generalizes by robustness and exact analogy but needs a softmax readout to interpolate; and (iv) geometric attention recovers interpolation, preserves analogy, and uniquely encodes order; (v) against classical one-shot and trained baselines, the model's measured edge is storage density and composable algebraic structure, not raw capacity or robustness; (vi) versor position binding provably generalizes RoPE and matches it in trained transformers once the frequency spectrum is designed; and (vii) under sequential knowledge editing, the frozen-model versor memory surpasses a rank-one ROME baseline on all three canonical metrics at 60 edits. Establishing whether these advantages persist on pretrained language models, real data, and the standard editing benchmarks is the natural next step.

Reproducibility. A self-contained implementation of $\mathrm{Cl}(n, 0)$, the versor memory, the sparse memory, the generalization battery, the geometric attention layer, and the JAX training code for E6–E7 (transformers, position mechanisms, the ROME baseline, and the versor editor)—together with the JSON logs behind every number reported here—accompanies this paper and is archived at doi.org/10.5281/zenodo.21752907.

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